

A. C. BRIDGES

Introduction

- Measurement of L , C , f
- Null method- very accurate

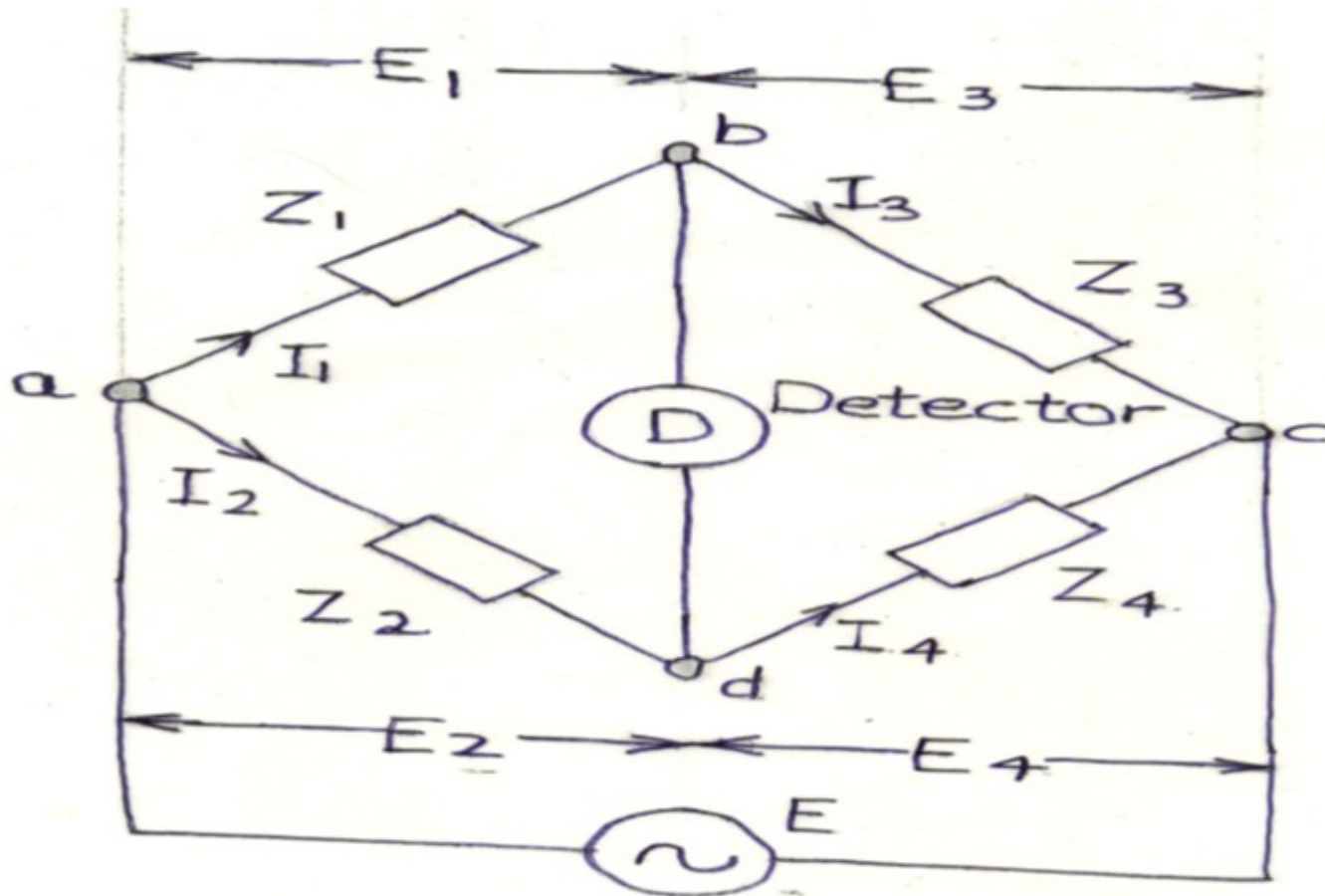
Sources

- Power Line- for low f
- Oscillators- for high f

Detectors

- Head phones: 250 Hz to 4 KHz
- VG: 5 Hz to 1000 Hz. Mostly used till 200 Hz.
- Tunable amplifier detectors: 10 Hz to 100 KHz. Very versatile.

Basic AC Bridge



For balance

$$\vec{Z}_1 \vec{Z}_4 = \vec{Z}_2 \vec{Z}_3$$

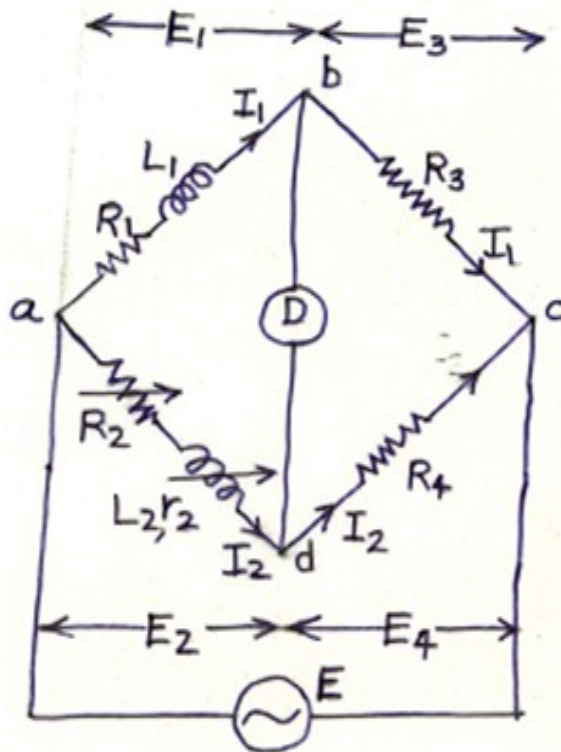
i.e. Magnitudes & phases must match

$$Z_1 Z_4 / \theta_1 + \theta_4 = Z_2 Z_3 / \theta_2 + \theta_3$$

* Real and imaginary parts should be separately equal.

Measurement of Self Inductance

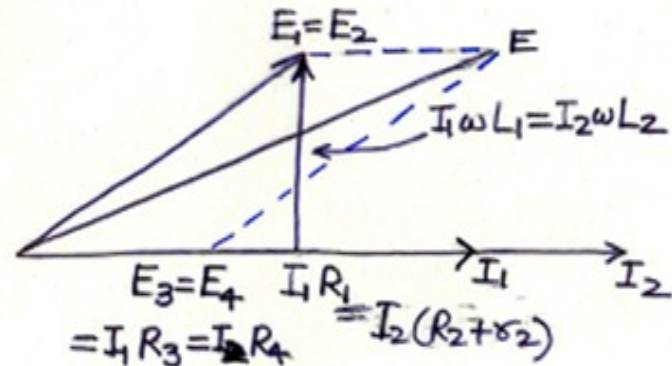
Maxwell's Inductance Bridge



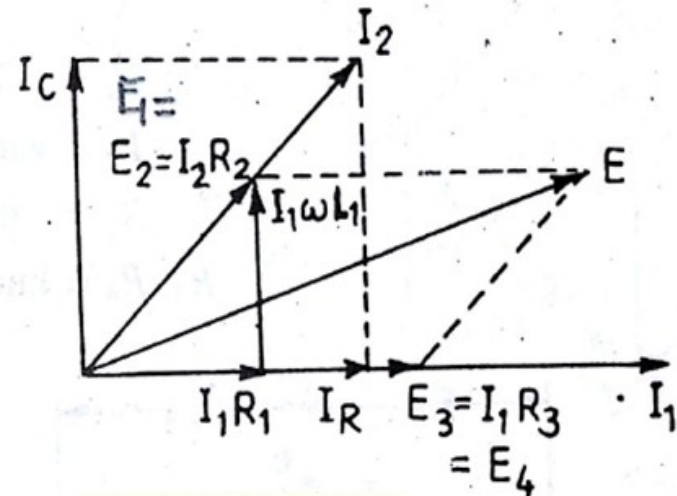
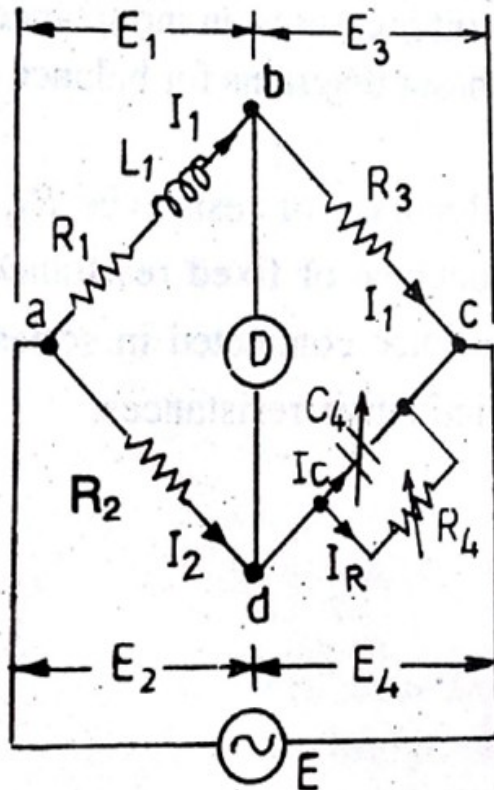
At balance

$$L_1 = \frac{R_3}{R_4} L_2$$

$$R_1 = \frac{R_3}{R_4} (R_2 + r_2)$$



Maxwell's Inductance Capacitance Bridge



$$R_1 = \frac{R_2 R_3}{R_4}$$

$$L_1 = R_2 R_3 C_4$$

$$= \frac{I_C}{\omega C_4} = I_R R_4$$

Adv. $\rightarrow$ f doesn't appear in equations

Dis. $\rightarrow$ (1) C is costly.

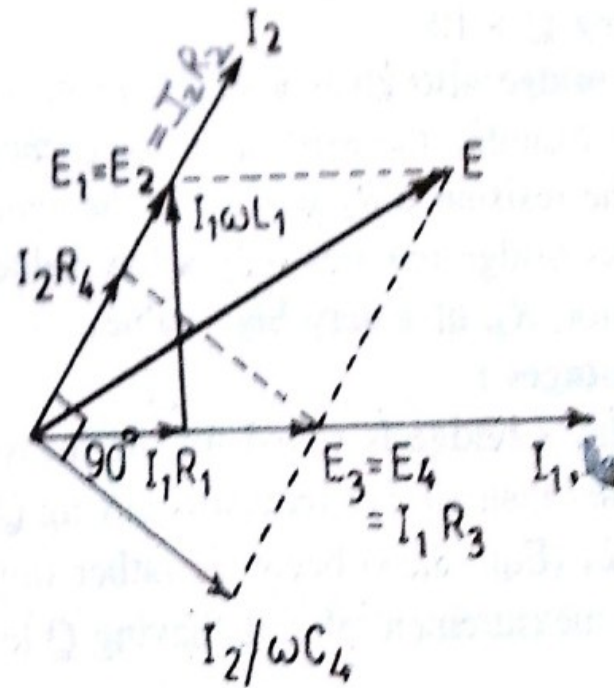
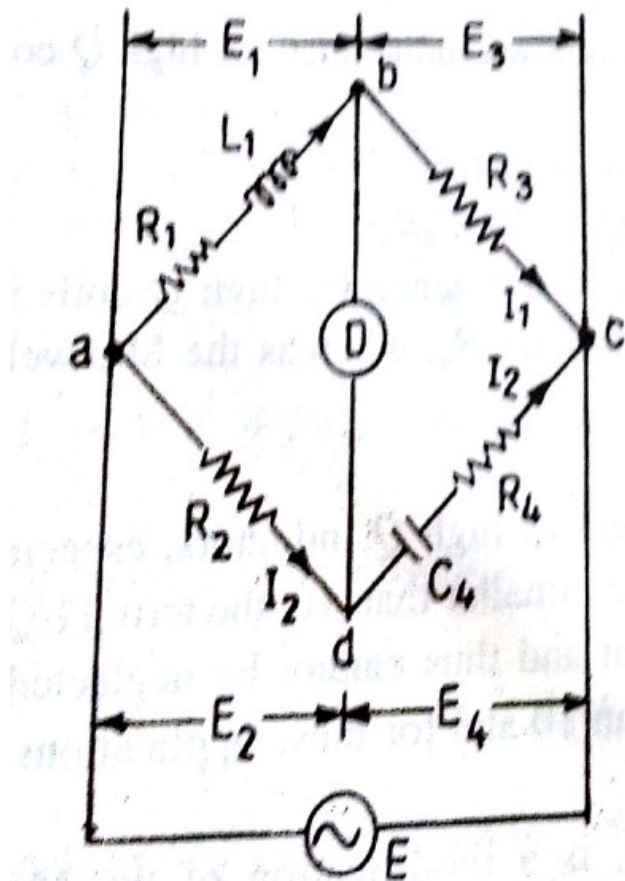
(2) Bridge is limited to measurement of ^{medium} ~~low~~ Q coils. ($1 < Q < 10$). ^{Used for} medium Q coils

$$Q = \frac{\omega L_1}{R_1} = \omega C_4 R_4.$$

Measurement of high $Q \Rightarrow$ large $R_4 \Rightarrow$ high cost.

Measurement of low $Q \Rightarrow$ inductive resistors $\Rightarrow$ problem ⁱⁿ balance.

Hay's Bridge



$$L_1 = \frac{R_2 R_3 C_4}{1 + (\omega C_4 R_4)^2}$$

$$R_1 = \frac{\omega^2 R_2 R_3 R_4 C_4^2}{1 + (\omega C_4 R_4)^2}$$

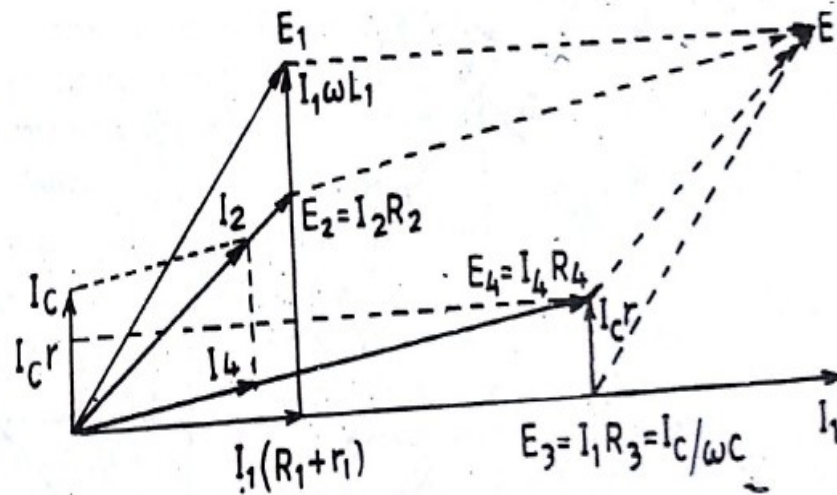
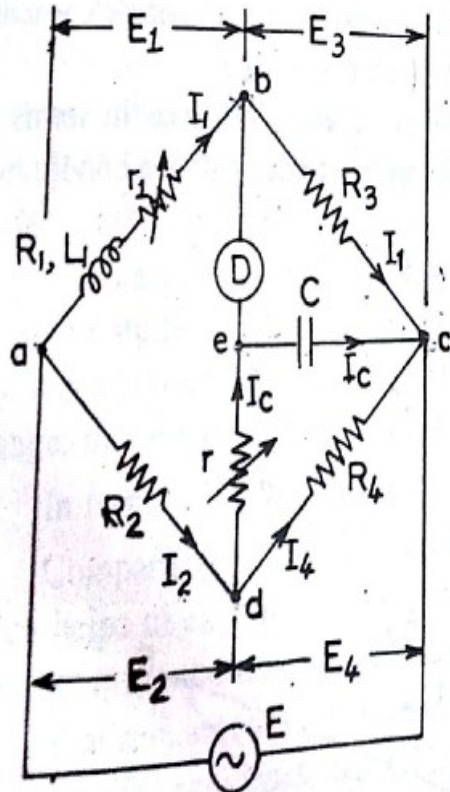
$$Q = \frac{\omega L_1}{R_1} = \frac{1}{\omega C_4 R_4}$$

For a high Q coil

$$\therefore L_1 = \frac{R_2 R_3 C_4}{1 + (1/Q)^2} \Rightarrow L_1 = \frac{R_2 R_3 C_4}{1} = R_2 R_3 C_4$$

* Used for coils having $Q > 10$.

Anderson's Bridge



Self inductance is measured in terms of a standard capacitor. This method is applicable for precise measurement of self inductance over a very wide range of values.

*For balance 'b' & 'c' should be equipotential

$$I_1 R_3 = \frac{I_c}{j\omega C}$$

$$I_1 (r_1 + R_1 + j\omega L_1) = I_2 R_2 + I_c r$$

$$I_c \left(r + \frac{1}{j\omega C} \right) = I_4 R_4$$

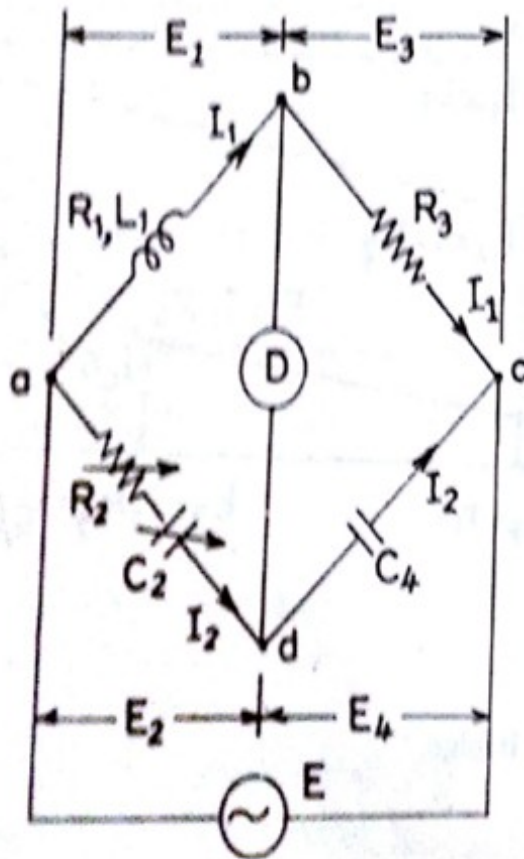
At balance

$$R_1 = \frac{R_2 R_3}{R_4} - r_1$$

$$L_1 = c \frac{R_3}{R_4} [r(R_4 + R_2) + R_2 R_4]$$

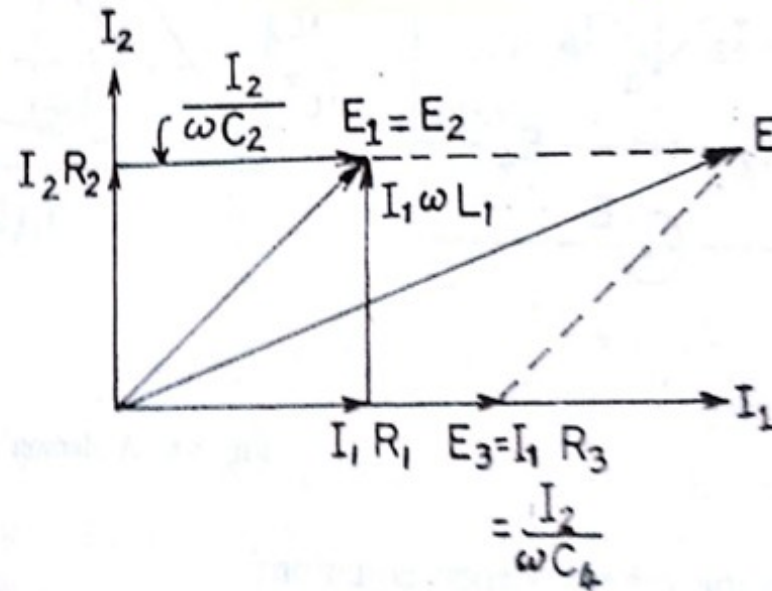
- * For easy convergence of balance, alternate adjustments of x_1 & x should be done.
- * Especially useful for low Q coils.
- * Structure is complex.

Owen's Bridge



$$L_1 = R_2 R_3 C_4$$

$$R_1 = R_3 \frac{C_4}{C_2}$$

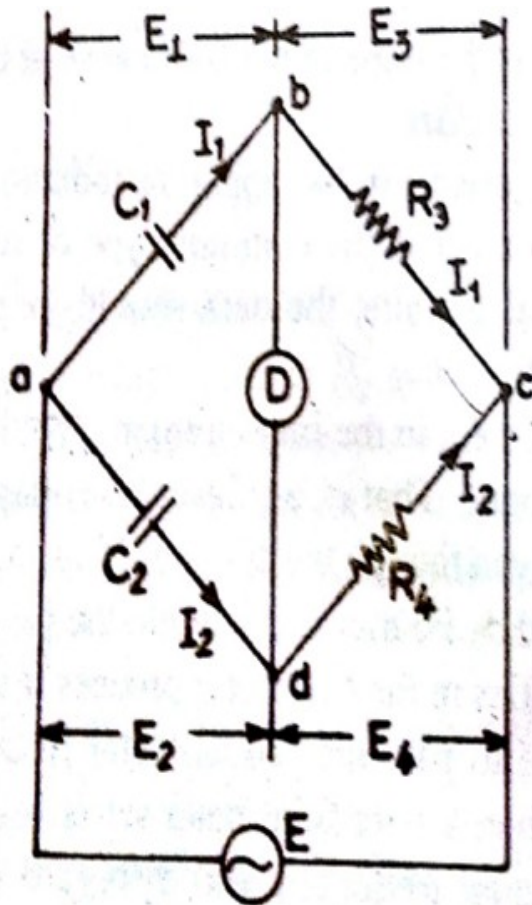


* Balance is obtained easily and bridge can be used over a wide range for measurement of inductances.

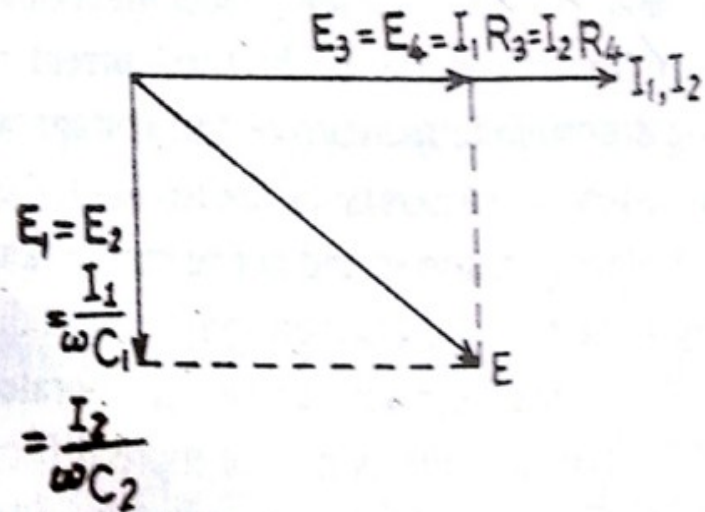
* Bridge requires a variable capacitor which is very expensive.

Measurement of Capacitance

DeSauty's Bridge

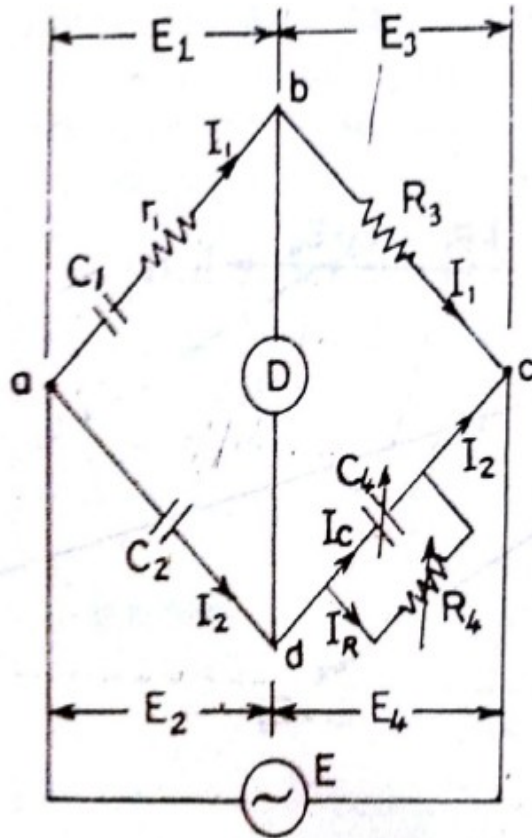


$$C_1 = C_2 \frac{R_4}{R_3}$$



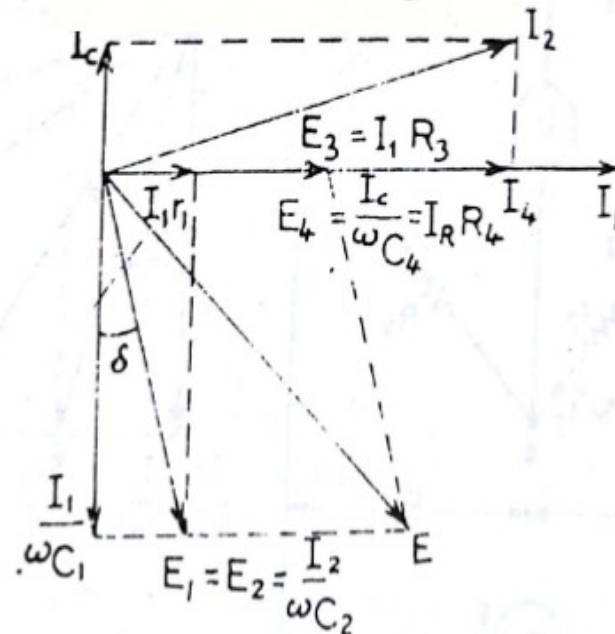
- * Very simple in construction.
- * Balance obtained by varying either R_3 or R_4 .
- * Method can be applied only for lossless capacitors.

Schering Bridge



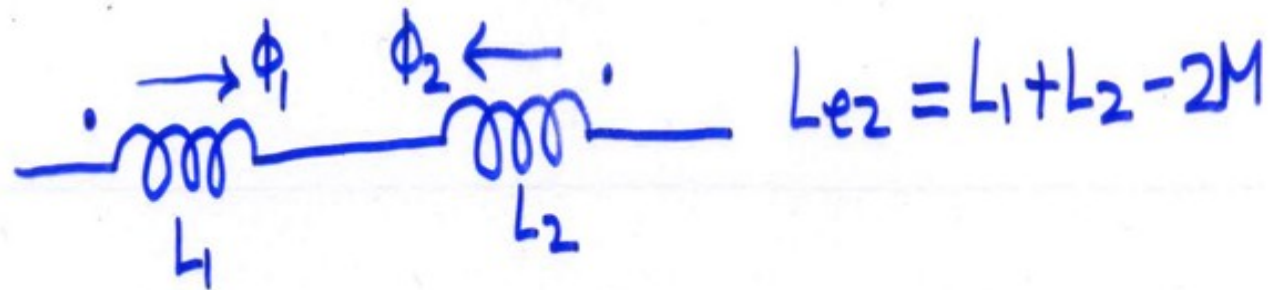
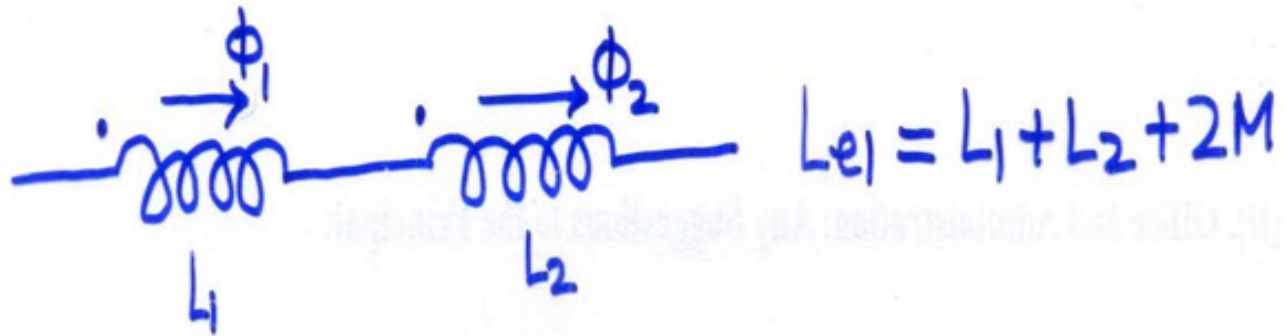
$$\delta_1 = R_3 \frac{C_4}{C_2}$$

$$C_1 = C_2 \frac{R_4}{R_3}$$



Measurement of Mutual Inductance

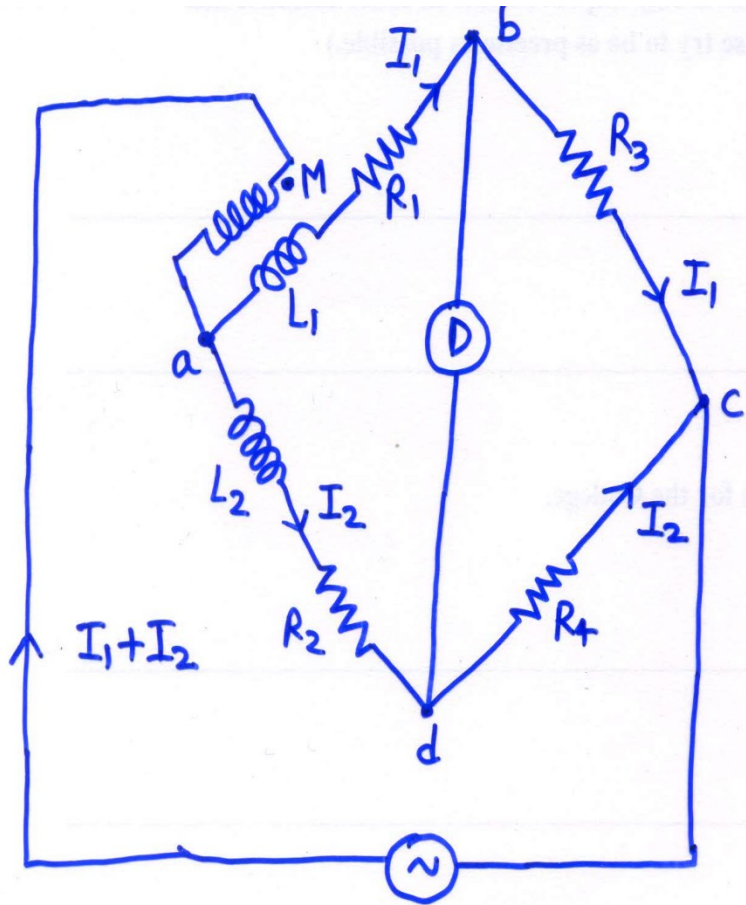
Basic Method



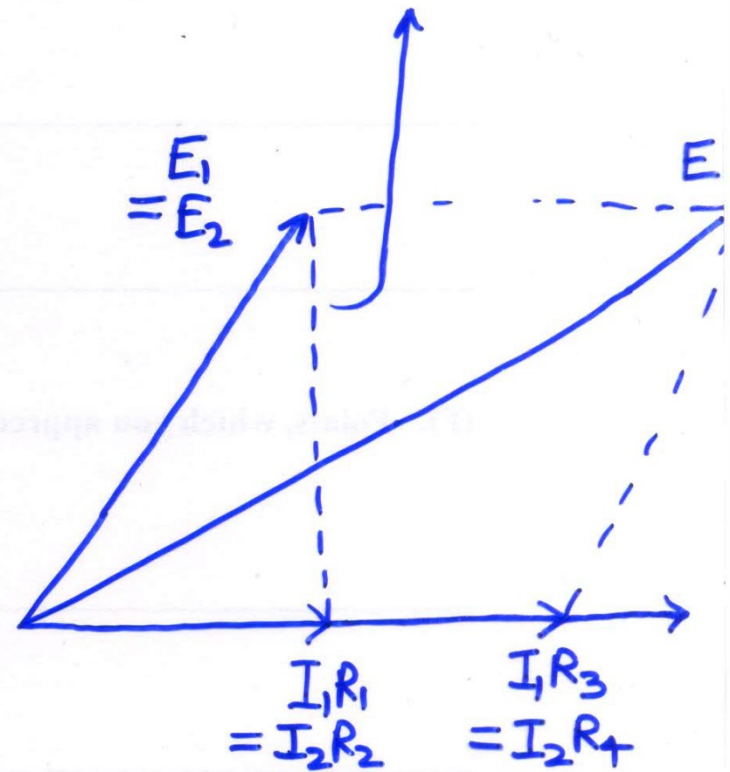
$$M = \frac{1}{4} (L_{e1} - L_{e2})$$

Connect to an AC bridge to find L_{e1} and L_{e2} . M can then be computed by the adjacent formula.

Heaviside Bridge



$$I_2 \omega L_2 = (I_1 + I_2) \omega M + I_1 \omega L_1$$



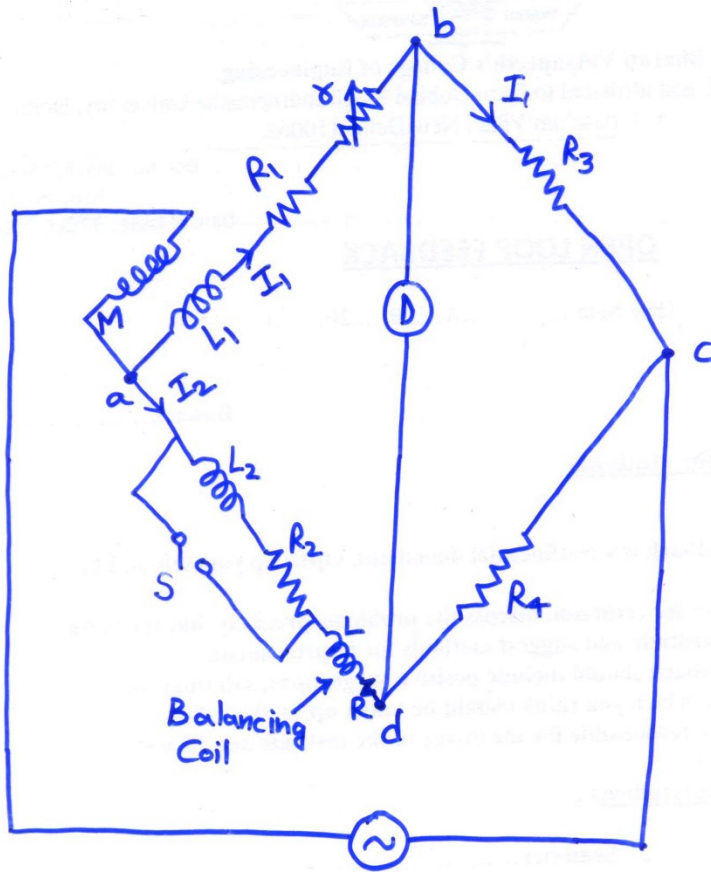
At balance: $I_1 R_3 = I_2 R_4$

& $(I_1 + I_2)(j\omega M) + I_1(R_1 + R_3 + j\omega L_1) = I_2(R_2 + R_4 + j\omega L_2)$

$$R_1 = \frac{R_2 R_3}{R_4} \quad M = \frac{R_3 L_2 - L_1 R_4}{R_3 + R_4}$$

Campbell's Modification

- Measures L in terms of M .



With switch open $L_2 + L = \frac{M_1(R_3 + R_4) + R_4 L_1}{R_3}$ & $R_2 + R = \frac{(R_1 + r_1)R_4}{R_3}$

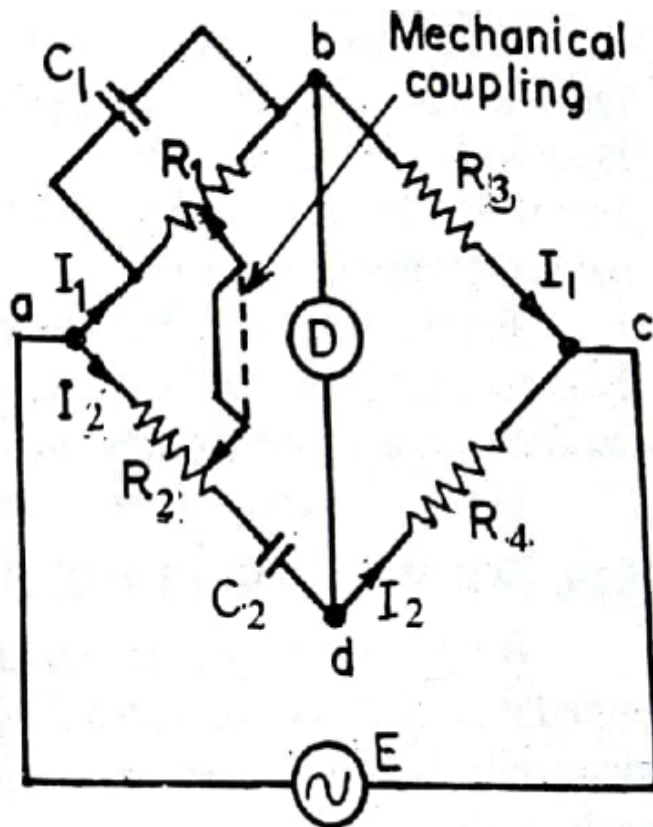
With switch closed $L = \frac{M_2(R_3 + R_4) + R_4 L_1}{R_3}$ & $R = \frac{(R_1 + r_2)R_4}{R_3}$

$$\therefore L_2 = (M_1 - M_2) \left(1 + \frac{R_4}{R_3}\right)$$

$$\& R_2 = (r_1 - r_2) \frac{R_4}{R_3}$$

Measurement of Frequency

Wien Bridge



At balance

$$\left(\frac{R_1}{1 + j\omega C_1 R_1} \right) R_4 = \left(R_2 - \frac{j}{\omega C_2} \right) R_3$$

Equating the real and imaginary parts

$$\omega C_1 R_2 - \frac{1}{\omega C_2 R_1} = 0$$

$$\text{from which } \omega = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}$$